could discover a way of spreading themselves that did not depend on the conventional sperm or egg route, they would take it and be less cooperative. This is because they would stand to gain by a different set of future outcomes from the other genes in the body. We've already seen examples of genes that bias meiosis in their own favour. Perhaps there are also genes that have broken out of the sperm/egg 'proper channels' altogether and pioneered a sideways route.

There are fragments of DNA that are not incorporated in chromosomes but float freely and multiply in the fluid contents of cells, especially bacterial cells. They go under various names such as viroids or plasmids. A plasmid is even smaller than a virus, and it normally consists of only a few genes. Some plasmids are capable of splicing themselves seamlessly into a chromosome. So smooth is the splice that you can't see the join: the plasmid is indistinguishable from any other part of the chromosome. The same plasmids can also cut themselves out again. This ability of DNA to cut and splice, to jump in and out of chromosomes at the drop of a hat, is one of the more exciting facts that have come to light since the first edition of this book was published. Indeed the recent evidence on plasmids can be seen as beautiful supporting evidence for the conjectures near bottom of page 237 (which seemed a bit wild at the time). From some points of view it does not really matter whether these fragments originated as invading parasites or breakaway rebels. Their likely behaviour will be the same. I shall talk about a breakaway fragment in order to emphasize my point.

Consider a rebel stretch of human DNA that is capable of snipping itself out of its chromosome, floating freely in the cell, perhaps multiplying itself up into many copies, and then splicing itself into another chromosome. What unorthodox alternative routes into the future could such a rebel replicator exploit? We are losing cells continually from our skin; much of the dust in our houses consists of our sloughed-off cells. We must be breathing in one another's cells all the time. If you draw your fingernail across the inside of your mouth it will come away with hundreds of living cells. The kisses and caresses of lovers must transfer multitudes of cells both ways. A stretch of rebel DNA could hitch a ride in any of these cells. If genes could discover a chink of an unorthodox route through to another body (alongside, or instead of, the orthodox sperm or egg route), we must expect natural selection to favour their opportunism and improve it. As for the precise methods that they